My Body, My Data

Designing Privacy-Preserving Period Tracking

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Prototype repository: https://github.com/majamishevska/3pt

1 Introduction

In 2025, users alleged that *Flo Period & Ovulation Tracker* shared sensitive menstrual and pregnancy data with third-party technology companies, leading Google and *Flo* to agree to a combined $56 million settlement [10]. This was not a niche case: *Flo* currently reports 77 million monthly active users and 420 million downloads [2]. It was also not a new concern. In 2019, reporting found that *Flo* had informed Facebook when users were having their periods or had logged an intention to get pregnant, despite saying that user data would not be shared with third parties [5]. *Flo* is not the exception so much as a visible example of a broader problem.

Period-tracking apps can reveal sexual activity, pregnancy status, fertility, symptoms, medication use, and intimate bodily patterns. After the *Dobbs* decision in the United States, choices about how this data is collected, stored, shared, or deleted are also choices about user safety, consent, autonomy, and power. This project asks what a responsible period-tracking app should look like when privacy, transparency, and user control are treated as design requirements rather than afterthoughts. We answer this through our report and prototype, *3PT* (*Privacy-Preserving Period Tracker*).

The report first maps out the problem space in Section 2, including prior work on menstrual app privacy, post-*Roe* data concerns, and best practices for responsible femtech design. Section 2.2 explains the course concepts that shaped our design stance, especially privacy as power, the limits of consent, transparency, and infrastructure. Section 3 presents input from reproductive health advocates, prior cycle-tracking design work, privacy research, and survey respondents. We then describe our artifact, justify the main technical choices behind it, and discuss the limits of what our prototype can solve.

2 Problem Space and Design Stance

2.1 Literature Review

Prior work shows that menstrual app privacy practices often fail to treat reproductive data with the sensitivity it deserves. Shipp and Blasco's systematic analysis of 30 menstrual apps found that privacy policies often completely disregarded reproductive-related data, even when that data was necessary for the app to function [14]. Hammond and Burdon similarly argue that menstrual tracking apps create both control harms, such as weak consent and secondary data use, and deeper intimate harms, where users' reproductive and sexual lives become exposed to systems built around data extraction–while reinforcing normative gender stereotypes and women as "reproductive citizens" [4]. In other words, the problem is not only that users may not read privacy policies, but it is that the policies and app architectures often do not give users meaningful control over their bodily data in the first place.

This issue became even more urgent after the overturning of *Roe v. Wade*. In a 183-participant survey, Cao et al. found that U.S. users became more cautious about period-tracking apps and often felt powerless over how their reproductive data could be used [1]. At the same time, a University of Cambridge report describes menstrual tracking data as a "gold mine" for advertisers, with risks for employment, insurance, cyberstalking, and abortion access [17]. Because many period trackers are not covered by HIPAA like traditional medical records, this data can be useful to users while also becoming valuable, or dangerous, to companies, advertisers, data brokers, and legal institutions [11].

Feminist technology scholarship helps to further explain why the problem goes beyond a developer's technical implementation. In "Zuckerberg, get out of my uterus!," Healy argues that fertility apps reinscribe social control over the female body as a "newly-digitalized reproductive machine" by centering ovulation, pregnancy, motherhood, and a narrow idea of the "normal" cycle [5]. More recent work moves from critique toward design guidance, as Krishnamurti et al. argue that digital reproductive health tools should be built through stakeholder-inclusive teams, person-centered design, and reproductive health equity [7]. Nellore's femtech case study also emphasizes user-informed, privacy-aware app design, pointing to common design gaps around hidden privacy settings, unclear data deletion, limited export, and assumptions about user identity or reproductive goals [8].

These best-practice frameworks shaped our own approach: a responsible period-tracking app should not simply add a better privacy policy after the fact. Developers should build privacy, transparency, data deletion, export, local control, and inclusive language into the app from the beginning.

2.2 Theoretical Approach

The course readings helped us define what was actually at stake in the design problem. Period tracking goes beyond being a privacy issue in the narrow sense of hiding information; it is also about power, consent, transparency, trust, and the kinds of subjects an app encourages users to become.

2.2.1 Privacy as Power and Self-Determination. Véliz argues that privacy is not just an individual preference, but a condition for autonomy and democratic life. Her central claim is that data creates power: "Their power stems from us, from you, from your data" [18, p. 1]. This is especially relevant for reproductive health data because the information is both intimate and actionable. Whoever controls period-tracking data may gain power over how users are categorized, targeted, judged, or investigated.

This directly aligns with Rouvroy and Poullet's concept of "informational self-determination." They argue that privacy is a fundamental prerequisite for self-development and democracy [12]. If an app's basic architecture depends on collecting and sharing

intimate data without meaningful user control, the user has already lost power. Our design stance therefore prioritizes keeping reproductive data out of the surveillance economy entirely.

2.2.2 Consent, Nudging, and Their Limits. Solove's critique of privacy self-management shows why consent screens and privacy policies are not enough. He argues that privacy self-management takes refuge in consent, focusing on whether users agree to data practices rather than whether those practices are acceptable in the first place. The problem is that this model disregards structural and cognitive problems, and asks consent to do more than it can: it "does not provide people with meaningful control over their data" [16, p. 1880].

Simmons gives us a stricter account of valid consent: it must be intentional, knowing, and voluntary [15, p. 276]. His account of tacit consent shows that silence only counts as consent when people have a reasonable way to dissent without facing severe consequences [15, pp. 279–280]. This challenges the logic of apps that treat continued use as agreement, even when leaving means a user loses years of critical health records. Furthermore, Schmidt and Engelen's work on the ethics of nudging [13] highlights how platforms often use interface design to steer user behavior. Rather than using dark patterns to nudge users toward data sharing, our design ethical stance is to implement protective defaults, ensuring the path of least resistance is the safest one.

2.2.3 Black Boxes, Infrastructure, and Transparency. Pasquale's work on black-box systems frames transparency as intelligibility. Pasquale notes that while data is staggering in breadth, the most important algorithms remain "out of our reach, available only to insiders" [9, p. 191].

Hull adds that internet architecture functions as "architectural disciplinary power," nudging users toward market norms of self-optimization [6, p. 1]. For our project, this means responsible design must refuse an architecture that turns users into data sources or optimization projects. The app should make basic data flows intelligible without burdening the user with constant surveillance.

2.3 Design Stance & Core Tensions

Our design stance prioritizes structural privacy and data minimization to protect user agency. Drawing on Rouvroy and Poullet we argue that informational self-determination cannot be achieved by overburdening the user with complex privacy settings, as Solove critiques. Instead, privacy must be built directly into the infrastructure through protective defaults as we prioritize meaningful user power over platform incentives.

Taken together, these readings helped us define the core tensions of the project:

- **Usefulness vs. privacy:** Features that users value most (like long-term pattern analysis) traditionally require vast amounts of aggregated data, but more data increases privacy and legal risk.
- **Consent vs. consent fatigue:** A user may click "agree" or keep using an app out of medical necessity, but that does not mean they understand or accept every downstream data use. Relying solely on consent policies shifts responsibility onto individuals who lack real bargaining power.
- **Transparency vs. burden:** Users need clear information about their data, but transparency should not become another exhausting task users must constantly manage to stay safe.
- **Autonomy vs. paternalism:** Protective defaults can drastically reduce risk, but the app should not become entirely paternalistic. The design challenge is to make the safer option the default while still preserving meaningful user agency.
- **User needs vs. platform incentives:** Users want memory, prediction, and self-knowledge. Companies, advertisers, and data brokers see the same data as commercially valuable. Responsible design recognizes that these incentives are often in direct conflict.

3 Domain Expert and Stakeholder Input

After mapping the broader problem space, we gathered input from reproductive health advocates, a student researcher who had already studied cycle-tracking users, a privacy scholar studying reproductive health data, and survey respondents with direct experience using or avoiding these apps. We needed this range of perspectives because reproductive health data, beyond being a technical privacy issue, is also legal, political, medical, gendered, and deeply personal.

3.0.1 Reproductive Health Advocacy Perspective. Planned Parenthood Advocates (PPA) at Brown, a campus advocacy group focused on reproductive health, rights, education, and access, helped us frame period-tracking data as politically and socially sensitive. We spoke with several PPA members, including members of its executive board. Three insights were especially relevant:

- **Profound distrust of third-party data sharing.** In the current U.S. legal and political context around reproductive health, third-party sharing feels especially risky. The concern is not only that companies might sell or misuse data, but that reproductive health data could become accessible to actors far outside the user's original intent, including advertisers, data brokers, government actors, or legal institutions.
- **The gendered nature of period-tracking apps.** These apps often collect data from people whose bodies and reproductive choices are already subject to social, medical, and political scrutiny. This shaped our understanding of period tracking as connected to gendered surveillance and control, not just wellness tracking.
- **Overemphasis on pregnancy and fertility.** Many apps frame the user primarily through ovulation, pregnancy, or fertility. This can alienate users who track for other reasons, such as pain, irregular symptoms, mood changes, medication, or general self-knowledge.

Although PPA members were not technical experts, their concerns echoed the feminist technology literature we reviewed: period-tracking apps are not neutral tools. Their input reinforced that responsible design means protecting data while also questioning the gendered and reproductive assumptions built into the app.

3.0.2 Prior Cycle-Tracking Design Work. Audrey Chou, a senior at Brown University and a researcher in the Human-Computer Interaction Lab, had previously conducted qualitative interviews with 13 users and built a project around cycles and productivity. The

apps that came up most often in her interviews were *Clue* and *Flo*. Chou described a common frustration with existing apps: users felt a "black box" dynamic where they were giving data without receiving meaningful insight in return, connecting back to Pasquale's account. For some users, the issue was not data sharing in the abstract, but whether the app offered enough value to justify the data it collected—without that, the app could feel like a "glorified diary."

Chou also pointed us toward the difficulty of exporting data from existing apps and noted the importance of data portability as part of responsible design. She noticed that there was often no easy mass export option or API; when export was possible, it could require contacting the company, waiting for a response, and receiving files that were lengthy to obtain or partially malformed.

3.0.3 Privacy and Data Brokerage Research Perspective. Dr. Diana Freed's work on reproductive health data and the data brokerage landscape helped us understand how sensitive information can move beyond the original app context. Her paper, "Medical Data for Sale," examines how reproductive health information can circulate through data brokerage systems [3].

As an Assistant Professor of Computer Science and Data Science at Brown University, her perspective strengthened our concern that app-level privacy is only one piece of a larger ecosystem. Even if a period-tracking app is designed responsibly, other data sources can still reveal reproductive health information. For our artifact, this made data minimization especially important: the app should avoid adding more sensitive data to an already risky data environment.

3.0.4 Survey Respondents as Stakeholders. We also treated survey respondents as stakeholders with lived knowledge of period tracking. Their responses, discussed in Section 3.1, helped us understand how people navigate trade-offs between privacy, convenience, accuracy, health documentation, and political risk.

3.1 Survey Findings

We surveyed 38 respondents about period-tracking apps, privacy concerns, and expectations for safer reproductive health technologies. The main pattern was tension: respondents valued these apps for memory, planning, symptom tracking, and bodily self-knowledge, but often did not trust where their data went or who could access it. We used these findings to define what a responsible period-tracking tool would need to do differently.

3.1.1 The Privacy Paradox: Concern Does Not Necessarily Stop Use. A clear pattern in the survey was the privacy paradox: respondents cared about privacy, but many kept using health apps anyway. As Solove points out, there is often a gap between people's stated concern for privacy and their actual behavior, but this does not mean people do not care about privacy [16, p. 1886]. Out of 38 respondents, 22 respondents, or 57.9%, agreed or strongly agreed that they are concerned about privacy in health apps, and 28 respondents, or 73.7%, worried that health app data could be shared without their knowledge. Yet 25 respondents, or 65.8%, also agreed or strongly agreed that they still use health apps despite privacy concerns.

This paradox is especially important because *Flo*, discussed in the introduction for allegedly sharing sensitive reproductive health data with third parties, was the most commonly used app in our sample: 18 out of 31 respondents who named apps, or 58.1%, had used it. *Clue* was second, used by 9 respondents, or 29.0%. Users were not simply choosing between privacy and convenience; they were weighing risk against habit, useful features, and real health needs.

One respondent captured this trade-off clearly:

> "Though I wish I could stop using period tracking apps because of privacy concerns, I have had irregular and intense periods since I was 10 [...] Having all my data stored in one place is important for me to be able to understand patterns, spot irregularities, and advocate for myself at doctors' appointments. I have routinely had my menstrual concerns dismissed, so I need to be able to be specific."

This challenged the idea that the safest answer is simply not to track, or to rely only on memory, body awareness, or a notes app. For some users, tracking supports symptom management and medical self-advocacy. Our artifact therefore needed to preserve the value of tracking while reducing unnecessary privacy risks.

3.1.2 Reproductive Health Data Is Politically and Socially Sensitive. Respondents did not treat period-tracking data as ordinary wellness data. Out of 37 respondents, 18 respondents, or 48.6%, said the current legal, political, or cultural climate around reproductive health had affected how they feel about using period-tracking apps either "a great deal" or "somewhat." Half of respondents, 19 out of 38, also agreed or strongly agreed that period-tracking data is more sensitive than data from other health apps, such as step count, sleep, or fitness apps.

Open-ended responses made this concern more concrete. Respondents named worries about "data sharing that can be used for legal action," "data being shared with companies or government," and "abortion laws and how period tracking apps could surveil women's bodies and jeopardize women's rights." One respondent wrote:

> "While this doesn't affect me personally, I am concerned that conservative states may use data from period tracking apps to target and prosecute women who have had abortions."

For our artifact, this meant the app had to be designed around the possibility that reproductive health data could move into legal, governmental, or institutional contexts.

3.1.3 Why Users Stop or Switch Apps. When respondents stopped or switched period-tracking apps, the most common reason was data trust. Among the 22 respondents who answered this question, 9 respondents, or 40.9%, selected concerns about how data might be collected, stored, or shared. Other common reasons were too many ads, selected by 7 respondents, or 31.8%; concerns related to the broader legal, political, or cultural climate, selected by 6 respondents, or 27.3%; and unreliability or inaccuracy, selected by 5 respondents, or 22.7%.

These answers show that existing apps often fail on more than privacy. They can feel extractive, distracting, and inaccurate. Respondents mentioned "medical and scientifical accuracy, data sharing for targeted advertising," and one said their app was inaccurate "maybe 4-6 months out of the year." For our design, this meant that a responsible period-tracking app should not just add a clearer

privacy notice to the same product model. It should avoid ads, unnecessary third-party sharing, and overconfident predictions, while focusing on the tracking functions users actually need.

3.1.4 Users Want Transparency, Local Control, and Data Portability. Respondents gave concrete recommendations for safer period tracking: control over storage, sharing, identity, deletion, and export. One respondent wanted "an option to choose not to share data" and to know "where and how my data is shared at any time." Others asked for no sale to "third parties or the government," "no email or association to me as an individual," data stored "only locally," and "more parsable legal terms" before signing up.

These requests matched a broader lack of understanding and control. Out of 38 respondents, 29 respondents, or 76.3%, disagreed or strongly disagreed that they understand how period-tracking apps use the data they collect. Similarly, 30 respondents, or 78.9%, disagreed or strongly disagreed that they feel control over how their data is used. When asked what an app should show about data practices, respondents wanted direct answers: where data is stored, who has access, what is collected, and who receives it. As one respondent put it:

> "I would want complete transparency. Who receives my data? Where is it stored?"

Another linked transparency to autonomy:

> "I want all of it! I want it to tell me what data they have access to, and where [...] and full autonomy to delete data."

These findings shaped three design requirements: local or user-controlled storage, plain-language privacy explanations, and deletion/export controls that are easy to find rather than hidden.

4 Key Technical Choices and Justifications

This section summarizes the main implementation choices behind *3PT* and how they connect back to our research findings. The full codebase and local deployment instructions are included in the GitHub repository listed in Appendix A. The intended users are people who menstruate or track cycle-related symptoms and want a simple way to maintain personal health records without giving that data to a cloud-based platform.

4.1 Period and Cycle Logging

The app lets users log period ranges, flow strength, symptoms, mood, and notes. The prediction model is intentionally lightweight, treating entries as flexible ranges rather than assuming perfectly regular cycles. Operating alongside an "Irregular cycles mode," the app minimizes strong assumptions about what a cycle should look like. Rather than forcing users into a single "normal" template, the interface supports the practical value users requested—remembering dates, noticing patterns, and maintaining records for medical self-advocacy—without becoming overly prescriptive.

At this stage, the logging system is intentionally basic: medication, contraception, custom symptoms, and richer emotion options are not yet included. These could be added in future iterations while preserving local storage and data minimization.

4.2 Privacy-First Local Storage

Entries and settings are stored locally by default. There is no required account, backend, or cloud sync in the prototype. Instead of burying data practices in a lengthy privacy policy, the app avoids creating risky data flows in the first place. The Privacy area includes short dropdown explanations for common questions, making the basics visible without turning privacy into a burden.

The app also includes a "Delete All Data" control, acting on the principle of the right to be forgotten. Because the data stays local, the user has absolute authority over deletion. They do not have to submit a request to a company or blindly trust that a remote server complied. *3PT* additionally includes optional PIN protection for opening the app as a practical layer of protection for more everyday situations.

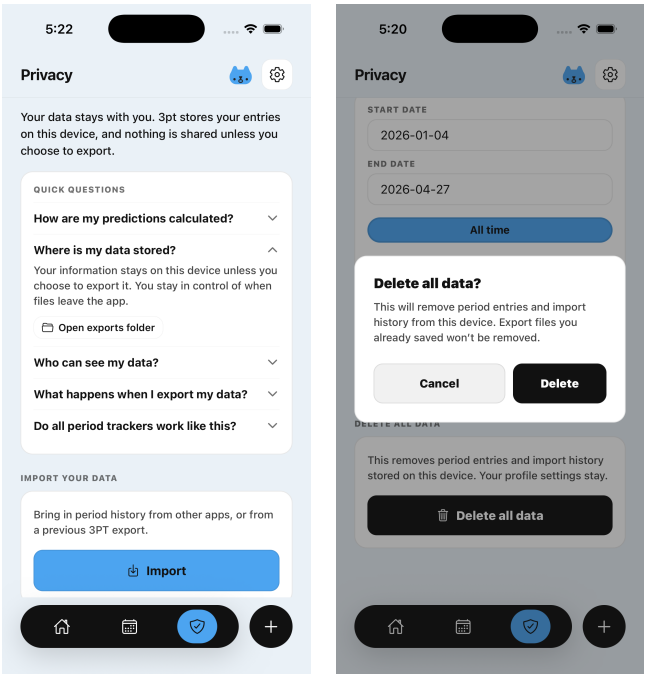


Figure 1: Privacy and deletion screens showing local-first storage, plain-language privacy explanations, and user-controlled deletion.

4.3 User-Controlled Import and Export

3PT supports export in JSON, CSV, and PDF formats. The PDF option is especially important for users who distrust digital systems, want an offline record, or may prefer to bring printed notes to a doctor's appointment. The app also supports imports from *Flo*, *Clue*, *Apple Health*, and previous *3PT* exports.

As our survey revealed, respondents had been using the same app for an average of 2.5 years. If users cannot move their own historical data, they lack true autonomy. Data portability ensures users are not locked in, as exporting should never be a trapdoor out of other platforms while our own system becomes another locked cage.

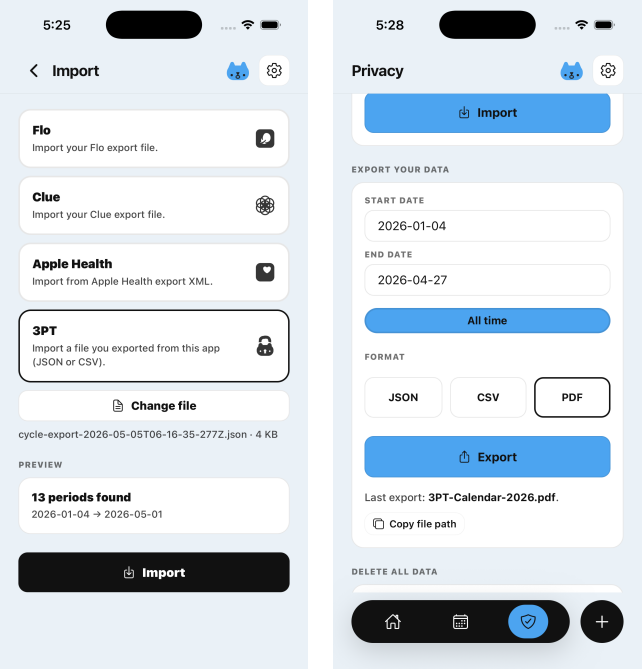


Figure 2: Import and export screens showing support for moving data into and out of 3PT.

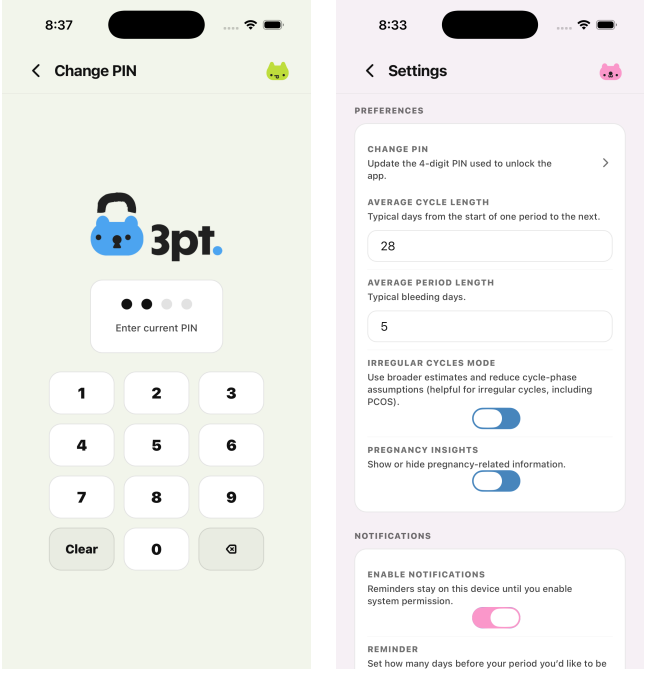


Figure 3: PIN and customization screens showing lightweight access protection and locally saved interface personalization.

4.4 Customization and Inclusive UX Controls

Because period tracking tools often embed rigid assumptions about gender, sexuality, and health, our interface deliberately avoids language that treats all users as women actively trying to conceive. Beyond a locally saved avatar customization feature, the app provides structural toggles like "Pregnancy insights" and an "Irregular cycles mode." Disabling pregnancy insights completely suppresses fertility-centered messaging, while the irregular mode suspends algorithmic assumptions about predictable cycle lengths. Drawing on feminist technology critiques—which note that apps do not just record bodies, but subtly dictate what kinds of bodies and goals are expected—our design refuses to push a single reproductive narrative as the default.

5 Limitations

Our artifact reduces some risks around period tracking, but it does not eliminate them.

First, local storage and PIN protection do not fully protect users if their device is compromised, searched, shared, or monitored. Encryption reduces the risk of exposed files or casual access, but if someone gains access to the unlocked device, the PIN, or the app itself, sensitive data can still be exposed. Privacy-preserving design can reduce data flows, but it cannot guarantee safety in every threat model.

Second, as Dr. Freed's insights help show, period data is only one part of a much larger reproductive health data ecosystem [3]. Even if a period-tracking app collects very little information, other data sources—such as location history, purchases, searches, and data broker profiles—can still reveal sensitive reproductive health information. Our artifact therefore addresses only one important site of risk, rather than the entire data landscape surrounding reproductive health.

Third, data minimization involves trade-offs. Collecting less data can limit prediction, personalization, reminders, or long-term pattern analysis. This is a deliberate trade-off in our design: we prioritize reducing exposure over maximizing functionality.

Fourth, plain-language transparency cannot solve every power imbalance between developers and users. Clear explanations can make the app easier to understand, but they do not fully remove the asymmetry between the people who design the system and the people who depend on it. More participatory design and user testing could help address this gap in future versions.

Fifth, as Chou pointed out, data portability also depends on other app developers. Our artifact can support export, but importing past data from existing apps depends on whether those apps make export possible and usable. A future extension could use local OCR or computer vision to turn screenshots or PDFs from other apps into importable files.

6 Conclusion

This project demonstrates that responsible period-tracking design requires fundamentally more than appended privacy policies and consent banners. In a sociopolitical landscape where reproductive health data is intimately tied to physical safety, legal risk, and commercial exploitation, responsible computing must treat privacy as an architectural imperative rather than a user preference.

Through our conversations with an advocacy group, researchers, and survey respondents, it became clear that users are often caught in a coercive tension: they need period-tracking tools for medical self-advocacy and bodily awareness, but they are forced to use platforms that treat their intimate data as commercial assets.

By grounding our technical artifact in the principles of data minimization, local-first storage, plain-language transparency, and robust data portability, we addressed the core anxieties surfaced by our stakeholders. While no software can fully insulate users from the broader ecosystem of data brokerage or compromised hardware, our approach proves that it is possible to build useful, empowering health tools that refuse the extractive norms of the surveillance economy. Ultimately, designing responsible technology means making explicit choices about what power should not be centralized, what data should never be collected, and what risks should never be shifted onto the user.

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A Prototype Submission Link

The prototype code is available on GitHub: **https://github.com/majamishevska/3pt**. Instructions for running and deploying the app locally are included in the repository README.

B Survey Materials

The survey included the following questions:

(1) What best describes your experience with period-tracking apps?
(2) Which apps have you used? Select all that apply.
(3) Approximately how long have you used a period-tracking app?
(4) If applicable, why did you stop using or switch period-tracking apps? Select all that apply.
    - The app felt unreliable or inaccurate
    - Too many ads
    - Features were not useful
    - Concerns about how my data might be collected, stored, or shared
    - Concerns related to the broader legal, political, or cultural climate
    - Better design/usability in another app
    - I no longer needed it
    - Other
(5) Has the current legal, political, or cultural climate, e.g. around reproductive health, affected how you feel about using period-tracking apps?
(6) Where do you think data from period-tracking apps is usually stored?
    - Only on my device
    - Mostly on company servers or in the cloud
    - Both on the device and on company servers / in the cloud
    - I am not sure
    - Other
(7) How confident are you in your answer above?

The survey also asked respondents to rate the following statements on a 1–5 scale:

(1) I am concerned about privacy in health apps in general.
(2) I worry that data collected by health apps could be shared without my knowledge.
(3) I trust period-tracking apps to handle my data responsibly.
(4) I feel I understand how period-tracking apps use the data they collect.
(5) I feel I have control over how my data is used in period-tracking apps.
(6) I still use health apps even when I have concerns about their privacy practices.

(7) I consider period-tracking data to be more sensitive than data collected by other health apps, such as step count, sleep, or fitness apps.
(8) I would be more concerned about the misuse of period-tracking data than the misuse of other kinds of personal data.
(9) I would be willing to give up some privacy in exchange for more features or convenience in a period-tracking app.

Finally, the survey included three open-ended reflection questions:

(1) What concerns, if any, do you have about using period-tracking apps?
(2) What would make you feel safer or more comfortable using a period-tracking app?
(3) What kinds of information would you want a period-tracking app to show you about how your data is collected, stored, used, or shared?
(4) Do you have any other thoughts, concerns, or ideas about period-tracking apps or your experience with them?